

Checkpoint 3

Data 6560 Sport Analytics
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Risk vs. Reward: Analyzing Aggressive vs. Conservative Decision-Making in Golf

Data Overview

The project relies primarily on shot level golf data to analyze decision making risk. The two main data sources under evaluation are the DataGolf API raw data and publicly aggregated PGA Tour datasets via Kaggle. These datasets provide detailed shot and performance level metrics needed to classify aggressive vs. conservative plays.

DataGolf

- **Origin:** Third party analytics platform partnered with PGA Tour; their data includes strokes gained metrics and shot details. datagolf.com
- **Collection Method:** They aggregate shot data, some via PGA Tour's ShotLink.
- **Access:** I have access to a paid account with them used in the past for sports betting models and odds

Kaggle PGA Tour Dataset

- **Origin:** Community published dataset on Kaggle covering PGA Tour seasons (2015–2022). [PGA Tour Data](https://www.kaggle.com/datasets/ahmedawad/pga-tour-data)
- **Contents:** Includes tournament and round stroke data, player performance, course data, strokes gained, etc.
- **Limitations:** Not necessarily raw shot by shot data; can be useful particularly with the strokes gained data

Data Structure & Key Variables

- **DataGolf:** JSON / API structure. Key variables include sg_app, sg_putt, sg_ott (strokes gained by category), prox_fw, prox_rgh, shot coordinates, shot counts, shot distance, lie type.
- **Kaggle PGA Tour:** CSV format, tournament statistics including score, par, strokes, player ID, round scores, and strokes gained variables.
- **Record Count & Coverage:** DataGolf covers many ShotLink equipped tournaments. Kaggle dataset covers multiple seasons.

Data Completeness & Consistency

There is some missing data and some duplicate data in both datasets, and even some events and full rounds that are missing strokes gained. There is also some inconsistent variable encoding between the datasets which will need to be fixed. Player ID will be used to ensure accuracy rather than player names

Some data quality issues and biases also exist. There is some sampling bias in the DataGolf data since it only covers ShotLink equipped tournaments. This may bias the results toward professional events, specifically ones with only the best players. There also may be some measurement error with ShotLink itself, where slight inaccuracy is recorded in location or proximity to the hole. This would be especially true for difficult lies, but shouldn't be significant enough to heavily impact the results. And lastly, there are some rounds missing SG (strokes gained) categories, which will have to be excluded from the analysis.

Initial Cleaning and Preparation

DataGolf has an API extraction using Python and the data_golf package to query and pull relevant shot data per tournament per player. Data will then be labelled as aggressive or conservative based on factors like lie, distance to green, and shot type. Missing SG categories will be identified and likely flagged and excluded from analyses. Variables will be standardized (lie types like fairway, rough, and sand will be consistently encoded), distances will be converted to a uniform unit (yards), and strokes gained metrics will be scaled if needed. The data sets may be linked if needed, which would involve merging the DataGolf data to the Kaggle dataset using player ID and tournament identifiers.

Conclusion:

Evaluating the quality and reliability of the shot data ensures the analysis will be credible and have minimal issues. Assessing coverage, consistency, and bias will allow subsequent modeling of risk and reward in shot making strategy.